Giacomo Antonioli

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Research Interests

Quantum numerical linear algebra: structured block encodings (semiseparable, hierarchically off-diagonal low-rank/HODLR) and tailored alternatives to Quantum Singular Value Transformation for matrix functions $f(A)b$, linear systems, and Krylov-type methods, with resource-aware circuits (depth, ancillas). Development of the quantum circuit simulators QCLAB (MATLAB) and QCLAB++ (C++), including distributed-memory simulation. Also quantum machine learning: variational circuits for state preparation and classification, and quantum orthogonal neural networks.

Education

2023 – 2027 **Ph.D., University of Pisa** — Quantum Computing Foundations. Expected Feb. 2027.
Thesis title: *Quantum Linear Algebra: From Matrix Representations to Structured Algorithms and Computational Realization.*
Supervisors: Dr. Gianna Maria Del Corso (University of Pisa), Dr. Roel Van Beeumen (LBNL)

2020 – 2023 **M.Sc. Computer Science, University of Pisa** — Artificial Intelligence.
Thesis title: *Quantum Sobel Edge Detection Based on Amplitude Encoding.*
Supervisor: Dr. Gianna Maria Del Corso (University of Pisa)
110/110 cum laude.

2016 – 2019 **B.Sc. Computer Science, University of Ferrara** — Computer Science.
Thesis title: *Estensione, Parallelizzazione e Applicazione alle Reti Neurali della Libreria Open Source per Calcolo Matriciale in Linguaggio Prolog* `ffimatrix`.
Supervisor: Dr. Marco Alberti (University of Ferrara)

Grants and Awards

2023 **PhD scholarship**, University of Pisa.
2025 **Competitive selection**, CEMRACS 2025 research session, fully funded.

Research Publications

Journal Articles

1. *Quantum block encoding for one-pair semiseparable matrices*
Antonioli, G., Boito, P., Del Corso, G. M., Porcelli, M.
Numerical Algorithms, 2026.

Conference Proceedings

1. *State Vector to Block Encoding: An Efficient Quantum Circuit Transformation*
Antonioli, G., Berti, A., Del Corso, G. M., Poggiali, A.
Proceedings of ICTCS 2026, CEUR Workshop Proceedings, vol. 4269, pp. 376–381, 2026.

2. *Outlier Detection and other applications of Quantum Matrix Multiplication*
Antonioli, G., Berti, A., Poggiali, A., Bernasconi, A., Del Corso, G. M.
2025 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), pp. 509–518, 2025.

3 *Variational Decision Trees with Structured Ansatzes*
Poggiali, A., **Antonioli**, **G.**, Berti, A., Del Corso, G. M., Guidotti, R.
2025 IEEE International Conference on Quantum Computing and Engineering (QCE), vol. 2, pp. 428–429, 2025.

4 *Variational Compression of Circuits for State Preparation*
Berti, A., **Antonioli**, **G.**, Bernasconi, A., Del Corso, G. M., Guidotti, R., Poggiali, A.
2024 IEEE International Conference on Quantum Computing and Engineering (QCE), vol. 02, pp. 44–48, 2024, DOI: 10.1109/QCE60285.2024.10250.

Preprints and Submitted

1 *Two-Tower quantum matrix chain multiplication: trading qubits for depth*
Antonioli, **G.**, Bernasconi, A., Berti, A., Del Corso, G. M., Poggiali, A.
arXiv:2607.13191, submitted to Future Generation Computer Systems, 2026.

2 *Block encoding of semiseparable matrices via their generators*
Antonioli, **G.**, Del Corso, G. M., Van Beeumen, R.
arXiv:2610.13228, 2026.

Technical Reports

1 *SQMS Quantum R&D in Machine Learning, Optimization and Sensing beyond Fundamental Physics Applications*
Venturelli, D., Lupascu, A., Dupont, M., *et al.*, 2024.

Software

2025 – now — **QCLAB / QCLAB++** — MATLAB and C++ quantum circuit simulators (with Dr. R. Van Beeumen, LBNL).
Implemented the distributed-memory (MPI) extension of QCLAB++, proposed by Dr. R. Van Beeumen, with OpenMP and GPU support.

Research & Professional Experience

Jul.–Aug. 2025 — **Researcher** — CEMRACS 2025, CIRM, Marseille, France
Selected through a competitive call (fully funded) for the 5-week research session on the THALES-sponsored project *"Quantum Group Convolution for PDEs,"* after the 1-week quantum computing summer school.
Supervisor: Dr. Frédéric Barbaresco (THALES)

Apr.–Jun. 2025 — **Visiting Researcher** — Lawrence Berkeley National Laboratory (LBNL)
Developed block encodings of semiseparable matrices via their generators (arXiv:2610.13228); worked on the distributed-memory extension of QCLAB++; studied quantum phase estimation with compressive sensing.
Supervisor: Dr. Roel Van Beeumen

Teaching and Mentoring

2026 — **Co-supervisor**, M.Sc. thesis *Widening the Bit: Extending QCLAB to Qudits and Generalizing State Preparation*, University of Pisa.

2023 – 2025 — **Teaching Assistant**, Introduction to Quantum Computing (M.Sc.), University of Pisa. Led the coding labs.

2022, 2024 — **Teaching Assistant**, Introduction to Programming (B.Sc.), University of Pisa.

Teaching and Mentoring (continued)

2019 — **Teaching Assistant**, Numerical Calculus (B.Sc.), University of Ferrara.

Research Projects

Quantum Subroutines for Matrix Functions & Activation Functions
INDAM–GNCS Project (CUP: E53C24001950001)
Quantum algorithms for matrix functions and nonlinear activation functions in quantum machine learning.
My Contribution: Designed and implemented quantum algorithms for the block encoding of *semiseparable matrices* utilizing efficient product methods. Investigated techniques to introduce nonlinearity into Quantum Orthogonal Neural Networks (QONNs) via classical activation functions, ancillary qubit entanglement, and post-selection-based weighting.

Talks

Sept. 2026 — *State Vector to Block Encoding: An Efficient Quantum Circuit Transformation.*
ICTCS 2026, Udine, Italy

May 2026 — *Efficient Encoding of Semiseparable Matrices in Quantum Circuits.*
ILAS 2026, Quantum NLA minisymposium, Virginia Tech, Blacksburg, VA, USA

Quantum Spectral Methods for Poisson and Heat Equations.
ILAS 2026, contributed talk, Virginia Tech, Blacksburg, VA, USA

Oct. 2025 — *Quantum Group Convolutions for PDEs and Equivariant Neural Networks.* (invited)
THALES Seminar on Quantum Algorithms & Computing, Paris, France

Nov. 2024 — *Variational Compression of Circuits for State Preparation.*
AIQxQIA Workshop, Bolzano, Italy

Outreach

Nov. 2025 — **Co-Organizer**, Pisa Quantum Festival, University of Pisa.

2025 — **Speaker**, Pisa Bright Night (European Researchers' Night): quantum computing for the public.

Nov. 2024 — **Co-Organizer**, Pisa Quantum Festival, University of Pisa; organized and led the hackathon.

Other Experience

2021 – 2023 — **System Architect & Division Leader**, E-Team Squadra Corse (University of Pisa Formula SAE team): led the Driverless division.

2020 — **Software Developer**, Bioretics Srl: 3D life-science imaging algorithms and ML monitoring tools.

Skills

Programming — C++ (C++17, MPI, OpenMP, GPU offload), MATLAB, Python, LaTeX.

Languages — Italian (native), English (C1).